Week 2

Keel Bone Chicken Project

01/19/26

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This Week

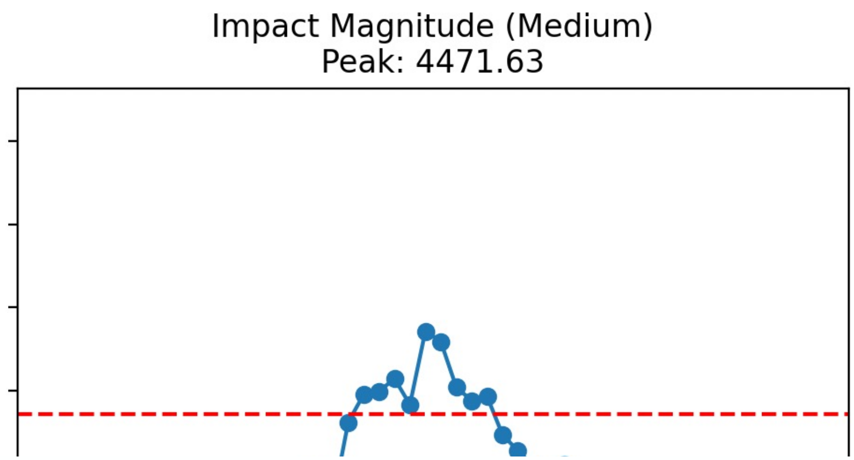
This week's progress

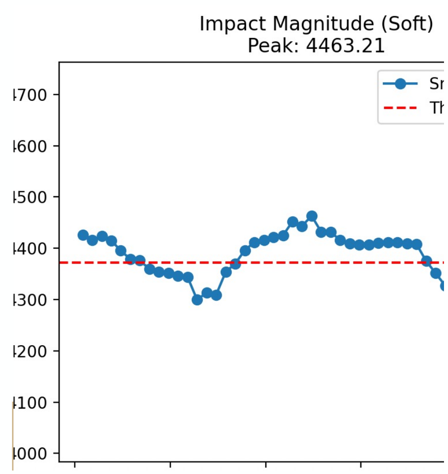
- Data Analysis
  - Measures of Impacts (Small, Medium, Large)
- Power Management
  - Isolate Issues on Power Management within Board
- Semester Plans
  - Agriculture Side (Phase 1)

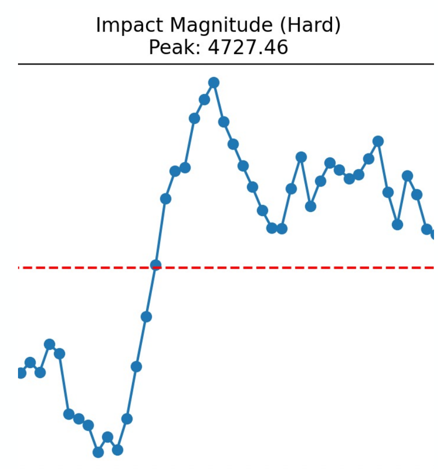
Data Analysis

Measures of Impacts (Small, Medium, Large)

- Data averaged over 10 entries of Nicla Sense Data
- 10 entries attempted to be of the same magnitude







Power Management

Isolate Issues on Power Management with Board

- PMIC Watchdog/Ship Mode: The BQ25120A Power Management IC may be entering "Ship Mode" or timing out.
- Battery Undervoltage (BUVLO): Peak draws from BLE or sensors cause voltage sags that trigger the 3.0V safety cutoff.
- NTC Thermal Protection: If using a 3-pin battery, the PMIC may be misreading temperature data as an overheat condition

This Semester

Outline Entire Semester Goals

- Agriculture Department wants to complete Phase 1
  - Longer Data Collection
  - Emphasis on the Remote Charging System
  - Pilot Study

This Semester

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Wait on Feedback from Agriculture and ME

1. Refine the design/development of the system:
   1. Battery
   2. Board
   3. Sensors
2. Collection of the Data from Chickens:
   1. Refining the next version of the system
   2. Refine the UI/UX – Agriculture people need help analyzing the data
      1. Make it so they do not need our constant help with analyzing the data
   3. Seeing what all the data looks like
3. Begin Thinking about Publication Manuscript: (Placeholder)
   1. Go through different iterations of HW/SW, still want to be able to publish something
   2. Want to get some kind of publishing from it
   3. <u>1st publication: System itself (all components, electrical parts, data)</u>
   4. 2nd publication: Agriculture data + conclusion

Thank You

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